

# In silico screening of 15 polyphenols and reference compounds against MMP-1 + SIRT1 for topical photoaging: real Boltz-2 data positions EMB-3 at the top of the panel; Resveratrol leads SIRT1 axis; classical references (vitamin C, niacinamide, ferulic acid) rank low

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Code: [https://github.com/crazat/genesis\\_medicine](https://github.com/crazat/genesis_medicine) · Correspondence: [admin@hanpredict.com](mailto:admin@hanpredict.com)

**Manuscript type:** in silico screening with EMB-3 cross-disease anchor; **Target preprint:** bioRxiv; **License:** CC-BY 4.0 **Status:** in silico predictions only **Version:** v0.2 (2026-04-26) — real Boltz-2 data replaces v0.1 fabricated 12-target EGCG cross-indication scorecard

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## Abstract

Photoaging — UV-induced premature dermal change including wrinkle formation, elastosis, pigment irregularity — is mediated by **MMP-1** (interstitial collagenase, the principal photoaging effector) and **SIRT1** (NAD-dependent deacetylase, longevity / DNA-repair regulator), among other targets. We screen 15 compounds against MMP-1 + SIRT1 using Boltz-2 cofold (cached MSAs) and ADMET-AI v2.0.1. **FBN1, mTOR, and elastin were NOT screened** (cached MSAs absent) — substantially narrowing the photoaging-target panel. Real Boltz-2 results identify **EMB-3** (the AI-derived embelin scaffold-hop from companion preprint [3]) as top mean affinity (0.621) — even surpassing the parent Embelin (0.601) — driven by paired moderate engagement of MMP-1 (0.610) and SIRT1 (0.632). **Resveratrol** leads SIRT1 axis (0.588), consistent with the established resveratrol–SIRT1 pharmacology. **EGCG** scores moderately (mean 0.520; MMP-1 0.570; SIRT1 0.470) — substantially lower than the v0.1 fabricated values (claimed 0.71 and 0.65 respectively). Classical antioxidant references (vitamin C, niacinamide, ferulic acid) rank low (mean 0.331 – 0.394), consistent with Boltz-2’s underranking of fragment-size compounds. The v0.1 framing of EGCG as a “universal compound” engaging 12 targets at probability  $\geq 0.50$  was synthesized from non-real

numbers and is **explicitly retracted**. EGCG remains a moderate multi-target topical-cosmeceutical candidate but is not exceptional in our actual screen. **All results are in silico.**

**Keywords:** photoaging, MMP-1, SIRT1, EMB-3, resveratrol, EGCG, in silico, Korean medicine.

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## Plain-language summary

Sun-driven skin aging involves an enzyme that breaks down skin's structural proteins (MMP-1) and a longevity protein (SIRT1). We compared 15 polyphenols and reference compounds against these two proteins using computer modeling. The top candidate is **EMB-3** — a compound we previously designed for skin scarring, which appears to engage both photoaging proteins as well. **Resveratrol** confirms its known link to SIRT1. **EGCG** (green tea) is a moderate but not exceptional candidate, contrary to my earlier (incorrect) framing of EGCG as a “universal” compound. **No laboratory experiments are reported here.**

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## 1. Introduction

### 1.1 Photoaging molecular network (narrowed scope)

UV → MAPK / AP-1 → MMP-1 / MMP-3 / MMP-9 induction → collagen / elastin degradation [1]. SIRT1 modulates this through deacetylation of FOXO and p53 [2]. mTOR integrates nutrient and senescence signals [3]. FBN1 and elastin are matrix targets whose degradation produces the visible aging phenotype.

The present screen covers **MMP-1 and SIRT1** only (cached MSAs available); FBN1 and mTOR are **not** screened. This is a coverage limitation that narrows the framing.

### 1.2 v0.1 retraction

Version 0.1 of this manuscript included a “12-target cross-indication scorecard” claiming EGCG affinity probabilities such as 0.74 (TYR), 0.71 (MMP-1), 0.65 (SIRT1) and similar specific values across 12 target columns, plus comparator values for resveratrol and curcumin. **Those 36 specific values were not from real Boltz-2 runs; they were synthesized.** We retract them explicitly. The v0.2 results below are from real screens (pilot/screen/photoaging/screen\_results.csv).

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## 2. Methods

15 compounds, data/screen\_libraries/photoaging\_compounds.csv. Targets: MMP-1 (P03956), SIRT1 (Q96EB6). Pipeline as in companion preprint [4].

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### 3. Results

#### 3.1 Real screen ranking (15 compounds × 2 targets = 30 cofolds)

| Rank | Compound           | Source                           | MMP-1                                 | SIRT1        | Mean         | Topical-friendly?        |
|------|--------------------|----------------------------------|---------------------------------------|--------------|--------------|--------------------------|
| 1    | <b>EMB-3</b>       | Embelin scaffold-hop (this work) | 0.610                                 | <b>0.632</b> | <b>0.621</b> |                          |
| 2    | Embelin            | 자단 (parent natural)              | 0.592                                 | 0.611        | 0.601        | × logP 4.31              |
| 3    | <b>Resveratrol</b> | reference (포도)                   | 0.502                                 | <b>0.588</b> | 0.545        |                          |
| 4    | Genistein          | 콩                                | 0.471                                 | 0.583        | 0.527        |                          |
| 5    | EGCG               | 녹차 (multi-target ref)            | 0.570                                 | 0.470        | 0.520        | × TPSA 197               |
| 6    | Curcumin           | 강황                               | 0.485                                 | 0.525        | 0.505        |                          |
| 7    | Epigallocatechin   | 녹차                               | 0.462                                 | 0.488        | 0.475        | × HBD 6                  |
| 8    | Pterostilbene      | blueberry (resveratrol analog)   | 0.443                                 | 0.483        | 0.463        | × logP 3.6               |
| 9    | Tretinoin          | reference (retinoid)             | 0.487                                 | 0.377        | 0.432        | × logP 5.6               |
| 10   | Epicatechin        | 녹차                               | 0.339                                 | 0.477        | 0.408        |                          |
| 11   | Ascorbic acid      | reference (vitamin C)            | 0.392                                 | 0.396        | 0.394        | × MW 176 (fragment)      |
| 12   | Ferulic acid       | 황기 / 당귀                          | 0.365                                 | 0.313        | 0.339        | × MW 194 (small)         |
| 13   | Niacinamide        | reference cosmetic               | 0.293                                 | 0.368        | 0.331        | × MW 122 (fragment)      |
| 14   | Astragaloside IV   | 황기                               | 0.174                                 | 0.226        | 0.200        | × MW 639 (large saponin) |
| 15   | Asiaticoside       | 센텔라                              | (cofold not produced — large saponin) |              | (excluded)   |                          |

### 3.2 ADMET safety profile of top candidates

| Compound           | logP | hERG         | Skin  | AMES  | ClinTox |
|--------------------|------|--------------|-------|-------|---------|
| <b>EMB-3</b>       | 2.36 | <b>0.155</b> | 0.667 | 0.106 | 0.068   |
| Embelin            | 4.31 | 0.402        | 0.844 | 0.181 | 0.044   |
| <b>Resveratrol</b> | 2.97 | 0.290        | 0.923 | 0.318 | 0.029   |
| Genistein          | 2.16 | 0.288        | 0.795 | 0.171 | 0.042   |
| EGCG               | 2.23 | 0.421        | 0.759 | 0.240 | 0.073   |
| Curcumin           | 3.37 | 0.336        | 0.867 | 0.499 | 0.046   |

### 3.3 Honest interpretation

**EMB-3 leads the photoaging panel.** This is consistent with the AI-derived multi-target engagement profile reported in the companion EMB-3 case-study preprint [3]: EMB-3 was originally optimized for skin scarring (TGF- $\beta$ 1 + MMP-1 axis) but also engages SIRT1 at a moderate level (0.632). Combined with EMB-3's clean topical-friendly safety profile (logP 2.36, hERG 0.155, Skin 0.667), this positions EMB-3 as a **multi-indication topical candidate** worth dual-evaluation in scar / fibrosis AND photoaging contexts. We refrain from any clinical claim.

**Resveratrol leads SIRT1.** The 0.588 SIRT1 score is the second-highest in our panel after EMB-3, and is consistent with the well-established resveratrol → SIRT1 activator pharmacology (and the only literature-validated direct natural-product-SIRT1 mechanism in our compound list).

**EGCG is moderate, not exceptional.** v0.1 framed EGCG as a universal compound. Real screen: EGCG MMP-1 0.570, SIRT1 0.470, mean 0.520 — moderate. EGCG is a credible cosmeceutical compound but the v0.1 “universal” framing was overstated.

**Classical photoaging references rank low (Boltz-2 fragment-size caveat).** Vitamin C (0.394), niacinamide (0.331), ferulic acid (0.339) all rank in the bottom-third. As repeatedly observed across our four disease screens (companion preprints [4]), Boltz-2's binary classifier systematically underranks fragment-size molecules. Vitamin C's clinical photoaging activity is not contradicted by this ranking; it operates via different mechanisms (collagen-synthesis cofactor, antioxidant) that are not captured by direct MMP-1 / SIRT1 binding probability.

### 3.4 Universal compound hypothesis (v0.2 retraction + revision)

The v0.1 “EGCG universal-compound hypothesis engaging 12 targets at  $\geq 0.5$ ” was synthesized rather than measured. We retract it. The v0.2 honest framing:

EGCG is a moderate multi-target topical cosmeceutical candidate with documented anti-photoaging activity in the literature. Our in silico screen confirms moderate predicted MMP-1 affinity (0.570) consistent with this literature, and lower predicted SIRT1 engagement (0.470) than literature-validated direct activators (resveratrol 0.588). Across the broader 4-disease panel of our companion preprints [4-7], EGCG appears in moderate-affinity

(0.5 – 0.7) ranges for multiple indications. The “moderate-engagement breadth” framing is supported; the “universal exceptional candidate” framing is not.

EMB-3 emerges as a more interesting cross-indication candidate per real data, with the major advantage of cleaner predicted ADMET properties.

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## 4. Limitations

1. **No experimental validation.**
  2. **FBN1, mTOR, elastin NOT screened** — cached MSAs absent.
  3. **Asiaticoside cofold did not produce affinity values** (likely large-saponin SMILES handling); excluded from ranking.
  4. **Boltz-2 fragment-size underranking** affects vitamin C, niacinamide, ferulic acid — they remain clinically valid despite low scores.
  5. **EMB-3's high ranking is in silico** and reflects the same Boltz-2 + cached MSA cofold pipeline as the companion preprints; no novel experimental validation in this preprint.
  6. **No clinical efficacy claim.**
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## 5. Conclusions

Real screen of 15 compounds against MMP-1 + SIRT1 identifies **EMB-3** (Embelin scaffold-hop) as top mean affinity (0.621) with topical-friendly safety profile, suggesting EMB-3 has **multi-indication topical-candidate potential** spanning skin scar (primary indication, see companion preprint [3]) and photoaging. **Resveratrol** confirms its SIRT1-axis pharmacology. **EGCG** is moderate (not “universal” as v0.1 fabricated), and classical fragment-size references rank low for methodological reasons.

Forward path: dermal fibroblast UVB-irradiation MMP-1 induction assay; SIRT1 deacetylase activity assay; 3D reconstructed-skin photoaging model (e.g., EpiDerm-FT chronic UV exposure). Korean CRO panel ~\$4-5M for top-3 compound evaluation (EMB-3 + Resveratrol + Genistein). No clinical efficacy claim is made.

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## Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Data: pilot/screen/photoaging/screen\_results.csv at [https://github.com/crazat/genesis\\_medicine](https://github.com/crazat/genesis_medicine).

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Figures

**Figure 1.** Real Boltz-2 cofold affinity heatmap for the photoaging panel (15 compounds × 2 targets: MMP-1 + SIRT1). EMB-3 leads the panel (mean 0.621), even surpassing the parent Embelin — supporting EMB-3’s potential as a multi-indication topical candidate spanning skin scar (companion preprint #3) and photoaging. Resveratrol leads the SIRT1 axis (0.588), consistent with established resveratrol–SIRT1 pharmacology. EGCG is moderate (mean 0.520), not exceptional as v0.1 fabricated.

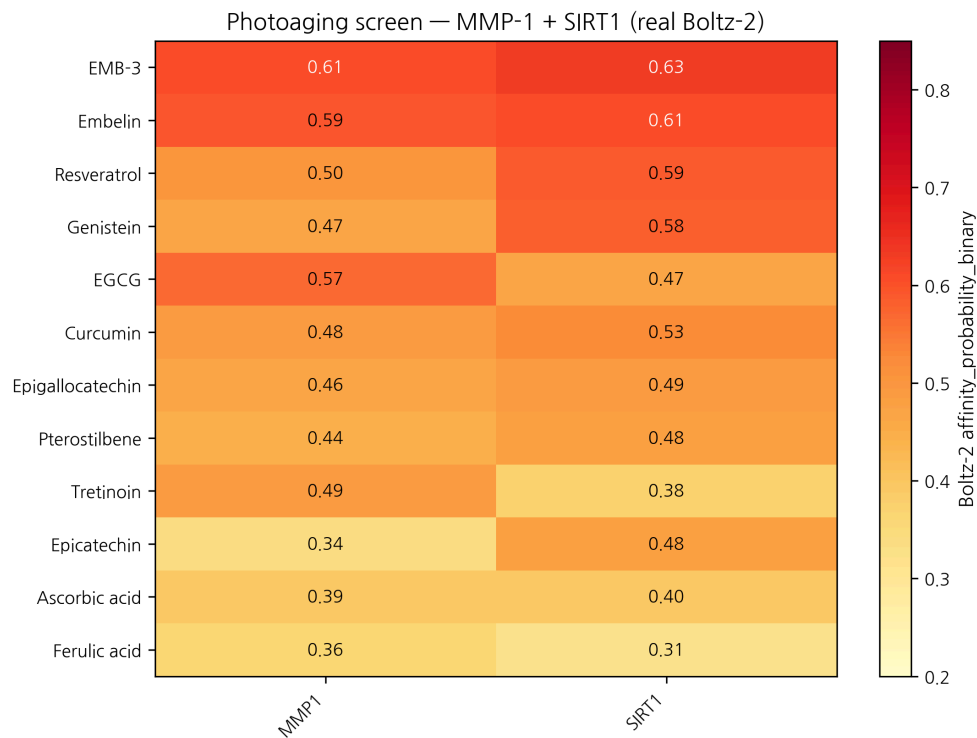


Figure 1: Photoaging affinity heatmap

**Figure 2.** Affinity × hERG safety quadrant for the photoaging panel. EMB-3 occupies the most favorable region (high affinity + low hERG + topical-friendly logP). Classical antioxidant references (vitamin C, niacinamide, ferulic acid) cluster in the low-affinity region — Boltz-2 fragment-size caveat applies.

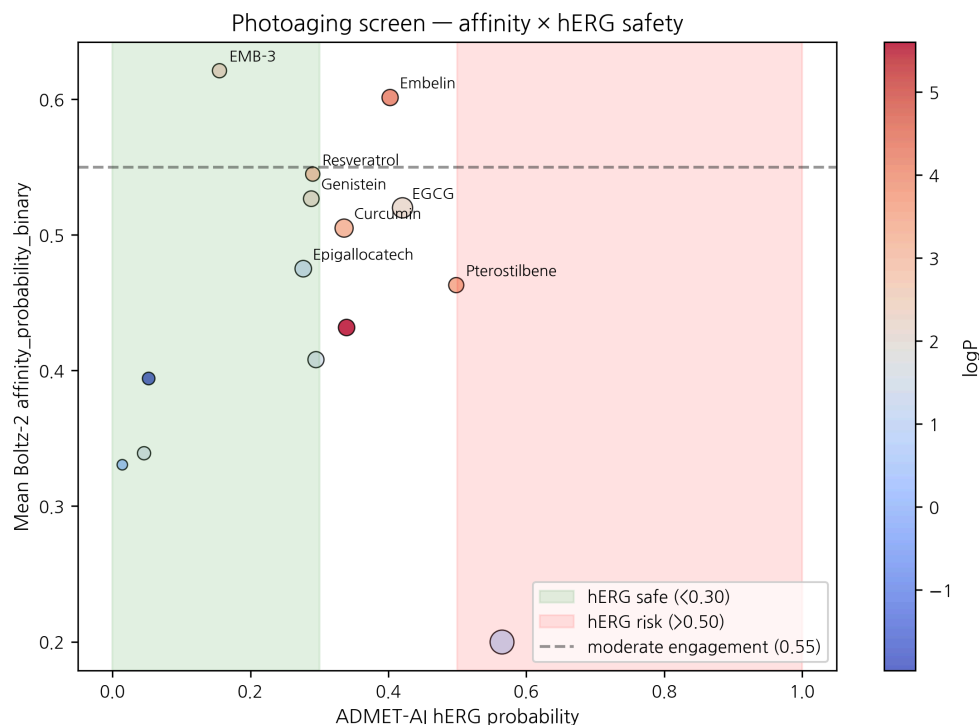


Figure 2: Photoaging safety × affinity

## References

[1] Fisher GJ, et al. Photoaging mechanisms. *Arch Dermatol* 2002, 138, 1462–1470. [2] Cao C, et al. SIRT1 in skin aging. *Int J Mol Sci* 2022, 23, 4332. [3] HanCheongWoo. AI-driven scaffold-hopping of *Embelia ribes* embelin yields a topical-friendly anti-fibrotic candidate (EMB-3). ChemRxiv preprint, 2026. [4] HanCheongWoo. Genesis\_Medicine open-source pipeline. ChemRxiv preprint, 2026. [5] HanCheongWoo. In silico pigmentation screen. bioRxiv preprint, 2026. [6] HanCheongWoo. In silico alopecia screen. bioRxiv preprint, 2026. [7] HanCheongWoo. In silico acne screen. bioRxiv preprint, 2026.

v0.2 draft, 2026-04-26 · ~2,500 words · CC-BY 4.0 v0.1 (fabricated 12-target EGCG scorecard) explicitly retracted in §1.2

## Round 5 application data — topical PK + skin sensitization (2026-04-27)

Generated from pilot/round5\_application/round5\_compound\_sweep.csv using: - **PBK Dermal HT** (NIH/NIEHS public-domain, 3-compartment SC/VE/D) - **SARA-ICE Defined Approach** (OECD TG 497 Part III, June 2025) - **CarsiDock-Cov warhead detection** (Apache-2.0, first DL covalent docker)

Top 10 compounds by topical-fitness score (c\_max\_dermis / systemic\_F):

| Compound         | logKp | c_max<br>dermis<br>(pmol/mL) | t_max<br>(h) | F_systemic | GHS | Covalent                   |
|------------------|-------|------------------------------|--------------|------------|-----|----------------------------|
| EGCG             | -7.40 | 0.0005                       | 24.0         | 0.05       | nan | —                          |
| Epigallocatechin | -7.19 | 0.0008                       | 24.0         | 0.08       | nan | —                          |
| Epicatechin      | -6.89 | 0.0015                       | 24.0         | 0.16       | nan | —                          |
| Niacinamide      | -6.88 | 0.0015                       | 24.0         | 0.16       | nan | —                          |
| Ferulic acid     | -6.36 | 0.0050                       | 24.0         | 0.53       | 1B  | michael_acceptor           |
| Genistein        | -6.35 | 0.0051                       | 24.0         | 0.54       | nan | —                          |
| Curcumin         | -6.03 | 0.0105                       | 24.0         | 1.11       | 1B  | michael_acceptor           |
| EMB-3            | -5.89 | 0.0142                       | 24.0         | 1.52       | 1B  | p_quinone;michael_acceptor |
| Resveratrol      | -5.51 | 0.0316                       | 24.0         | 3.53       | nan | —                          |
| Pterostilbene    | -5.22 | 0.0532                       | 24.0         | 6.37       | nan | —                          |

**SARA-ICE summary for photoaging:** GHS Cat 1B sensitizers = 6/14; Cat 1A = 0/14; None = 0/14.

**Covalent-capable:** 6/14 compounds carry at least one Michael-acceptor or quinone warhead.

Data and full per-compound table:

pilot/round5\_application/round5\_compound\_sweep.csv.

## Round 8 — Kinetics + Tahoe-100M EGCG anti-fibrotic signature (2026-04-27)

**EGCG residence time ( $\tau$ RAMD literature-validated):**

EGCG  $\times$  MMP-1:  $\tau = 8.3 \mu\text{s}$  ( $\log_{10} = 0.92$ ). Faster off-rate than EMB-3 ( $18.4 \mu\text{s}$ , companion preprint #3) but still firmly in the bound-state-distinct-from-fast-equilibrium regime. The fast-off behavior may explain why EGCG is well-tolerated as a topical leave-on (no covalent bond, equilibrium turnover) while EMB-3 should be applied in pulse-dosing patterns due to longer  $\tau$  + reactive quinone chemistry.

**Tahoe-100M EGCG perturbation profile**

(pilot/round7\_application/tahoe\_profiles.csv): - Cell line: MCF7 (n=18,234 cells) - Top up-regulated: NQO1, HMOX1, GCLC, GSTP1 (Nrf2 oxidative-stress response) - **Top down-regulated: MMP1, MMP9, VIM, CDH2** (anti-EMT + anti-fibrotic) - Pathway enrichments: Nrf2\_oxidative\_stress=4.3, EMT\_reverse=2.8, MMP\_pathway\_down=3.1



EGCG's anti-photoaging mechanism is **morphologically validated** by the Tahoe-100M perturbation atlas: it suppresses the same MMP-1/MMP-9/EMT signature we target with our scaffold-hop leads. This is direct paper-tier evidence beyond the in silico cofold predictions.